

## Lab 07 – 16-10-2023

**Task 1:** Write a program to find the roots of a quadratic equation using the formula:  $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ . Prompt the user for the input "a," accepting only non-zero values. Continuously request the user to re-enter the value until a non-zero value is provided. Additionally, check if the discriminant ( $b^2 - 4ac$ ) is negative. If it is negative, print "Sorry, the roots are imaginary." If the discriminant is non-negative, print both roots on separate lines. See sample output:

|                      |            |
|----------------------|------------|
| A: 0                 | A: 1       |
| Enter non-zero valu: | B: 5       |
| A: 0                 | C: 4       |
| Enter non-zero valu: | Root 1: -4 |
| A: -1                | Root 2: -1 |
| B: 1                 |            |
| C: 25                |            |
| Roots are imaginary  |            |

**Task 2:** Please input the runs and wickets for both team 1 and team 2. Ensure that all values are non-negative (zero is allowed). Additionally, check that if team 2 has more runs, their wickets should be less than 10.

After the input check is complete, if team 2 has more runs, print "**Team 2 has won by ... wickets**" (the number of remaining wickets for team B; if they have lost 6 wickets, there are 4 wickets remaining). If team 1 has more runs, print, "**Team 1 has won by ... runs**" (the difference in runs between team A and team B). In the event that both teams have the same runs but different numbers of wickets, the team with fewer wickets will win. Print "**Team 1 won by 1 run**" for team 1 and "**Team 2 won by ... wickets**" for team 2. Otherwise, print, "The match is a draw".

**Task 3:** Write a program to print two random numbers in a single line, until first number is smaller than second number. At the end, print **count** that is how many pairs generated.